Lecture-26
Internal Flows V: Minor losses in a pipe flow

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Learning Objectives

By the end of this lecture, you should be able to

1. Calculate local pressure drops in fittings and valves using empirical loss coefficients driven by **flow separation and turbulence.**
2. Evaluate head loss through **sudden or gradual geometric transitions.**
3. Convert **noncircular cross sections into an equivalent hydraulic diameter** to apply circular pipe friction and Reynolds number equations.

Minor losses in pipe flows

- Any piping system has a variety of fittings, bends, and **abrupt changes in area**, which cause **flow separation**.
- **Energy is dissipated** in these separated zones, and these are termed **minor losses**.
- At **inlets and exits**, flow separation leads to the formation of a **vena contracta**.

$$h_{l_m} = K\frac{\bar{V}^2}{2}$$

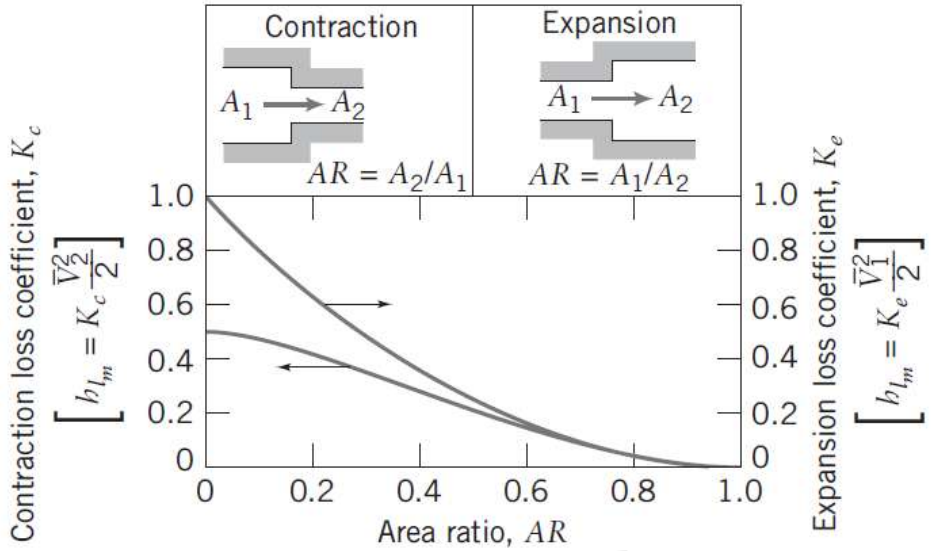
Entrance Type	Minor Loss Coefficient, K^a
Reentrant 	0.5–1.0 (depending on length of pipe entrance)
Square-edged 	0.5
Rounded 	r/D: 0.02 → K: 0.3; 0.06 → 0.2; ≥0.15 → 0.04

[a] Based on $h_{l_m} = K(\bar{V}^2/2)$, where $\bar{V}$ is the mean velocity in the pipe.
Source: Data from Reference [9].

Fox, Robert W., Alan T. McDonald, and John W. Mitchell. 2020. *Fox and McDonald's Introduction to Fluid Mechanics*. 10th ed. Hoboken, NJ: John Wiley & Sons.

Minor losses in pipe flows - continued

- At **enlargements and contractions**, we also encounter minor losses.
- The value of the loss coefficient depends on the **area ratio.**



	Included Angle, θ, Degrees						
A_2/A_1	10	15–40	50–60	90	120	150	180
0.50	0.05	0.05	0.06	0.12	0.18	0.24	0.26
0.25	0.05	0.04	0.07	0.17	0.27	0.35	0.41
0.10	0.05	0.05	0.08	0.19	0.29	0.37	0.43

Note: Coefficients are based on $h_{l_m} = K\bar{V}_2^2/2)$.
Source: Data from ASHRAE [9].

Fox, Robert W., Alan T. McDonald, and John W. Mitchell. 2020. *Fox and McDonald's Introduction to Fluid Mechanics*. 10th ed. Hoboken, NJ: John Wiley & Sons.

Minor losses in pipe bends and hydraulic diameter

- At **fittings and pipe bends**, we also encounter some minor losses.

Fitting	Geometry	K	Fitting	Geometry	K
90° elbow	Flanged regular	0.3	Globe valve	Open	10
	Flanged long radius	0.2	Angle valve	Open	5
	Threaded regular	1.5	Gate valve	Open	0.20
	Threaded long radius	0.7		75% open	1.10
	Miter	1.30		50% open	3.6
	Miter with vanes	0.20		25% open	28.8
45° Elbow	Threaded regular	0.4	Ball valve	Open	0.5
	Flanged long radius	0.2		1/3 closed	5.5
Tee, Straight through flow	Threaded	0.9		2/3 closed	200
	Flanged	0.2	Water meter		7
Tee, branching flow	Threaded	2.0	Coupling		0.08
	Flanged	1.0			

Source: Data from References [9] and [21]

- For noncircular pipes whose cross sections are **not too exaggerated**, we can calculate an **equivalent hydraulic diameter** and apply the relations for a circular pipe.

$$D_h \equiv \frac{4A}{P}$$

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Revisiting the learning objectives

1. Calculate local pressure drops in fittings and valves using empirical loss coefficients driven by **flow separation and turbulence.**
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